

Customer Shopping Behavior Analysis

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3,900 purchases show who buys, what they buy, and what moves them

Transactional data across four product categories

\$59.76

Average purchase

3.75

Average review rating

- Guide strategy on spending patterns, segments, product preferences and subscriptions



From raw data to analysis: Python, SQL, Power BI

01

Python

Loaded 3,900 rows and
18 columns

Filled 37 missing review
ratings with category
medians

02

Engineered fields

Age group and
purchase frequency

Dropped redundant
promo code column

03

PostgreSQL

Ten business questions
answered with SQL

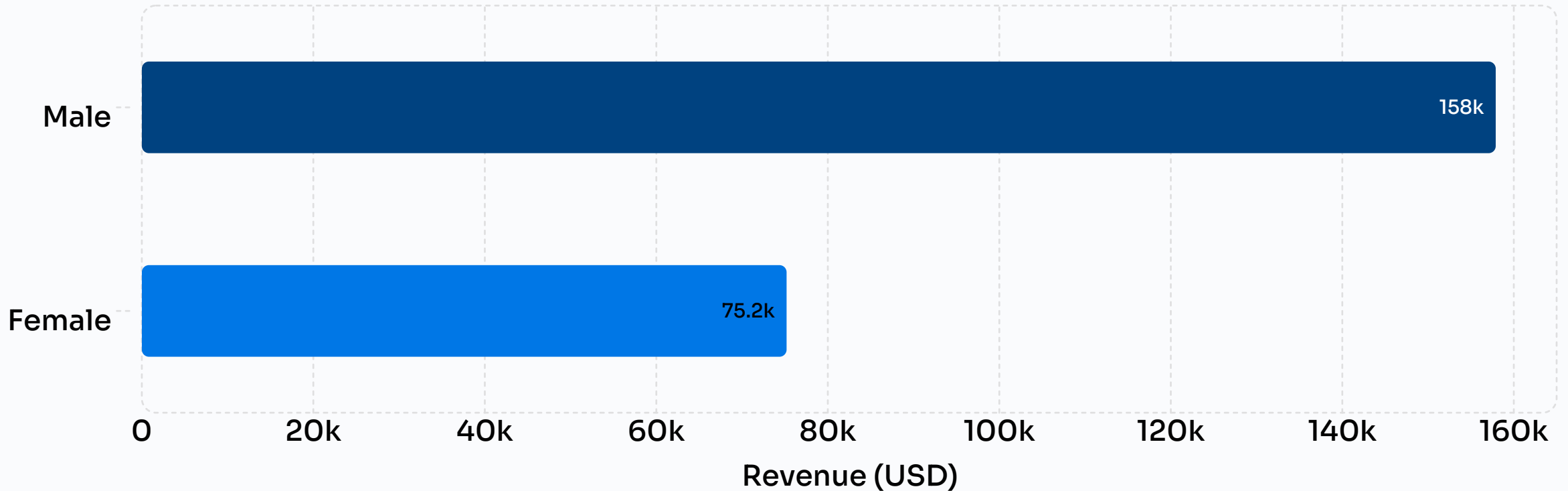
04

Power BI

Interactive dashboard
for stakeholders

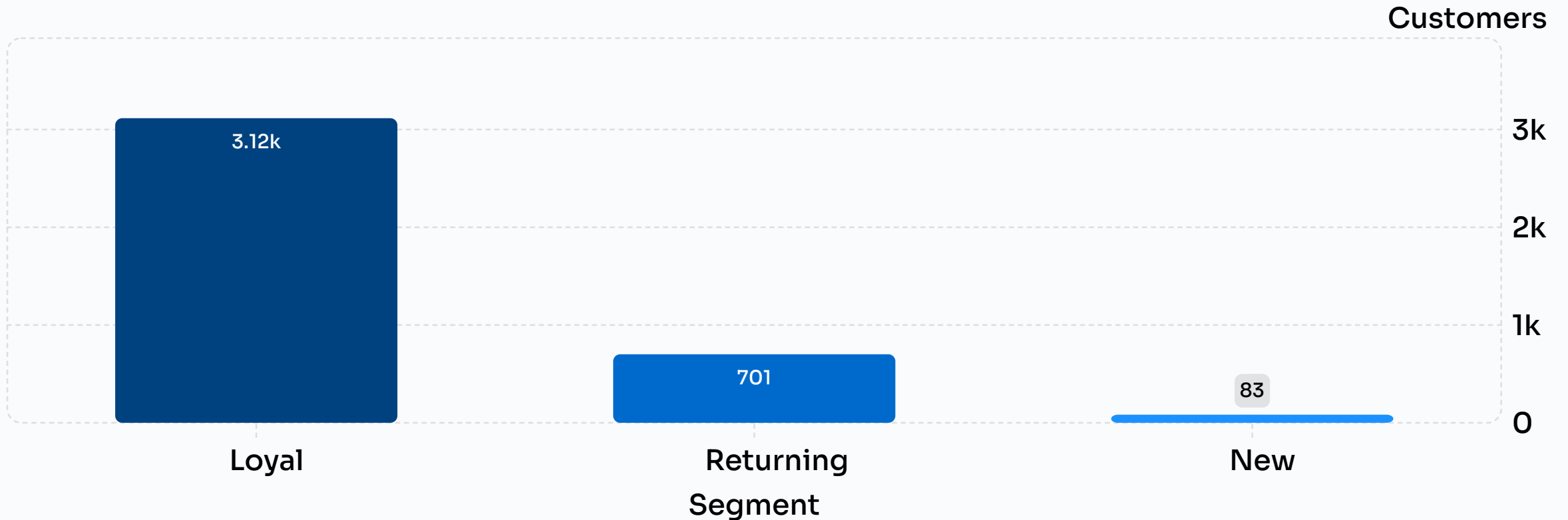
Male customers generate roughly twice the revenue of female customers

Gender



Male \$157,890 · Female \$75,191

Loyal customers dominate the base, and new customers are scarce



Loyal 3,116 · Returning 701 · New 83 customers

Subscribers spend the same as non-subscribers, but only 27% subscribe

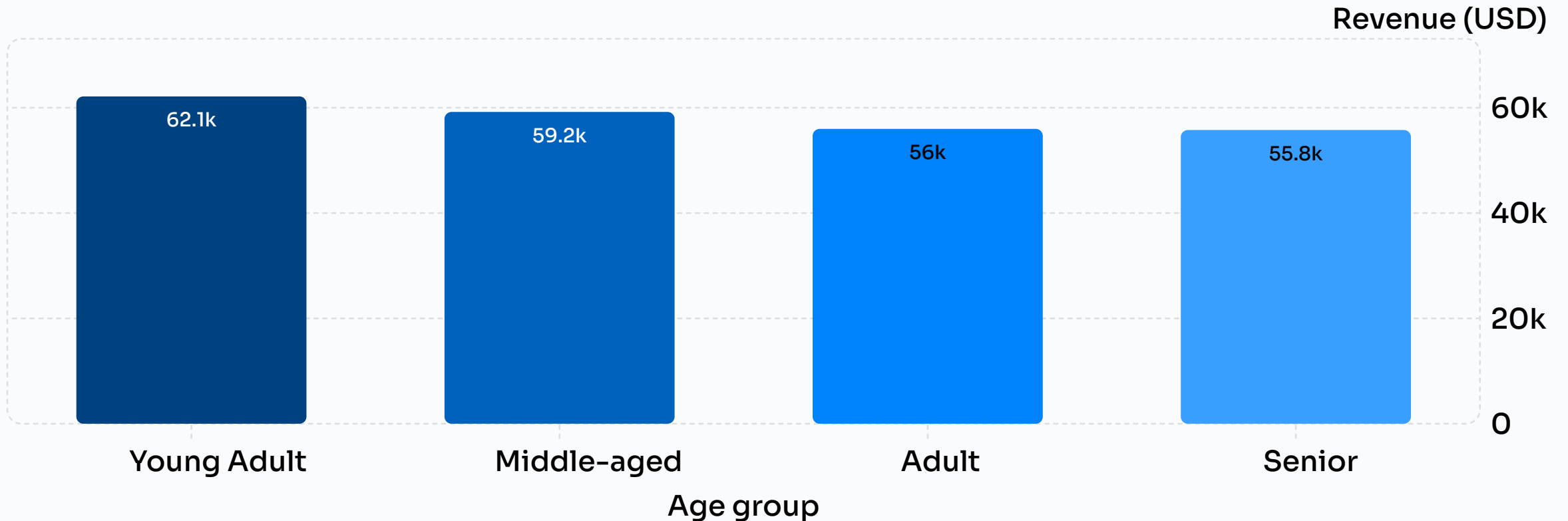
27%

Subscribe

	Subscribers	Non-subscribers
Customers	1,053	2,847
Average spend	\$59.49	\$59.87
Revenue	\$62,645	\$170,436

- Among repeat buyers with more than 5 purchases, 958 subscribe and 2,518 do not

Revenue is spread evenly across age groups, led by young adults



Young Adult \$62,143 · Middle-aged \$59,197 · Adult \$55,978 · Senior \$55,763

Best sellers differ from top-rated products

Top rated

Product	Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most ordered per category

Category	Product	Orders
Accessories	Jewelry	171
Clothing	Blouse	171
Footwear	Sandals	160
Outerwear	Jacket	163

Nearly half of purchases of some products rely on a discount

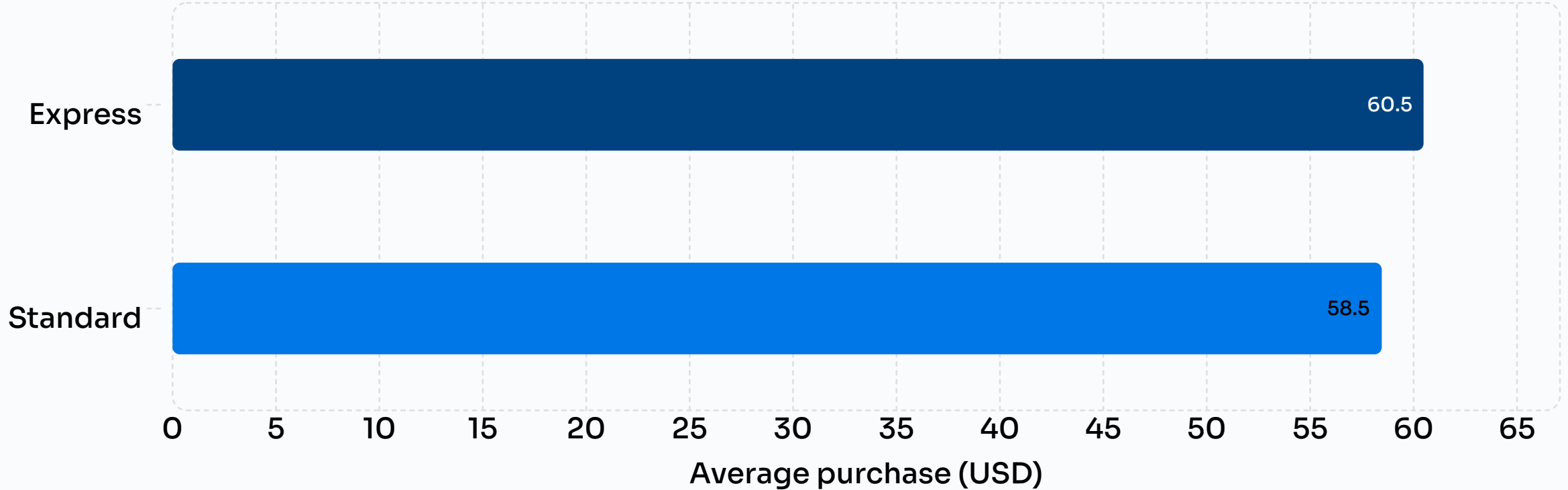
DISCOUNT RATE

50.00%	49.66%	49.07%	48.17%	47.37%
Hat	Sneakers	Coat	Sweater	Pants

839 customers used a discount and still spent above the average purchase

Express shipping buyers spend slightly more

Shipping type



Average purchase · Express \$60.48 · Standard \$58.46

Business Recommendations

Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the "Loyal" segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.

Suggestions

- **Subscriptions:** only 27% subscribe (1,053 vs 2,847) and spend is the same (\$59.49 vs \$59.87), so add subscriber value
- **Acquisition is thin:** 3,116 loyal customers vs only 83 new
- **Discount dependence is high** (Hat 50%, Sneakers 49.66%, Coat 49.07%): review discount policy for margin
- **Gender gap:** investigate male \$157,890 vs female \$75,191

Notes to take care of

- Express shipping buyers spend slightly more (\$60.48 vs \$58.46)
- Young adults lead age-group revenue (\$62,143)
- Data care: 37 missing review ratings were imputed with category medians